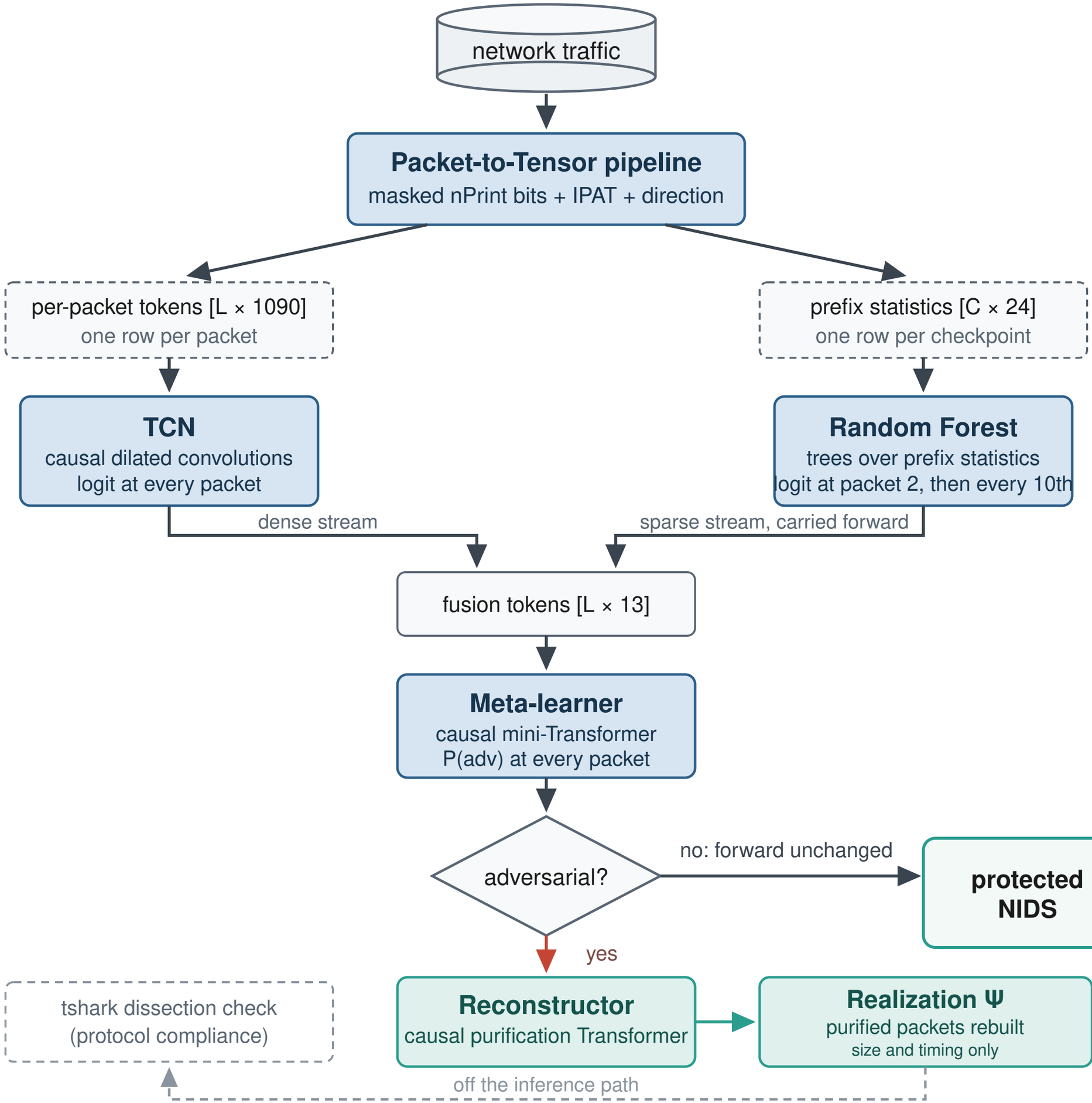
network traffic
Packet-to-Tensor pipeline
masked nPrint bits + IPAT + direction
per-packet tokens [L × 1090]
one row per packet
prefix statistics [C × 24]
one row per checkpoint
TCN
causal dilated convolutions
logit at every packet
Random Forest
trees over prefix statistics
logit at packet 2, then every 10th
dense stream
sparse stream, carried forward
fusion tokens [L × 13]
Meta-learner
causal mini-Transformer
P(adv) at every packet
adversarial?
no: forward unchanged
protected NIDS
yes
tshark dissection check
(protocol compliance)
Reconstructor
causal purification Transformer
Realization Ψ
purified packets rebuilt
size and timing only
off the inference path